CAMERA-LIDAR FUSION

Camera–LiDAR Sensor Fusion for Object Localization

RGB camera detects objects with YOLOv8; LiDAR provides 3D measurements.

Associate LiDAR points with detections to estimate object XYZ position and distance.

RGB camera → YOLOv8 → 2D detection

LiDAR → 3D measurements → point cloud

→

TF + projection

Fusion

→

Object XYZ

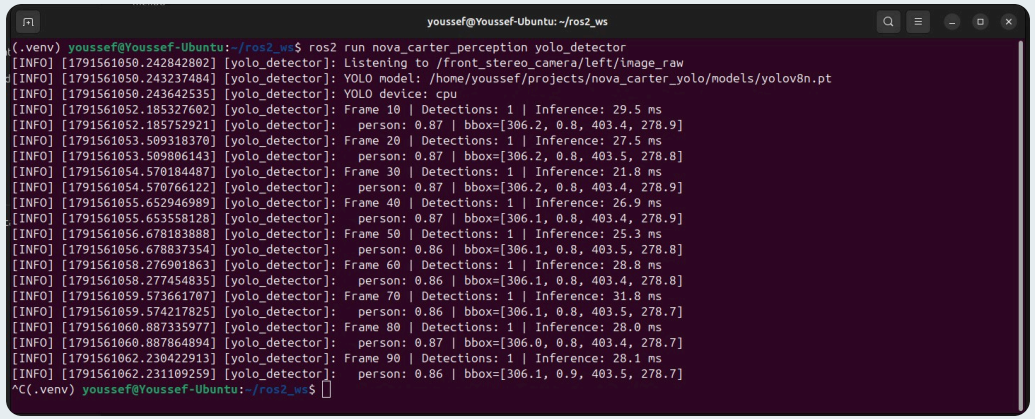
Distance

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CAMERA ONLY

YOLO knows what and where in the image, not how far

- ROS 2 camera stream from Isaac Sim
- YOLOv8n live person detection
- Class, confidence and 2D box

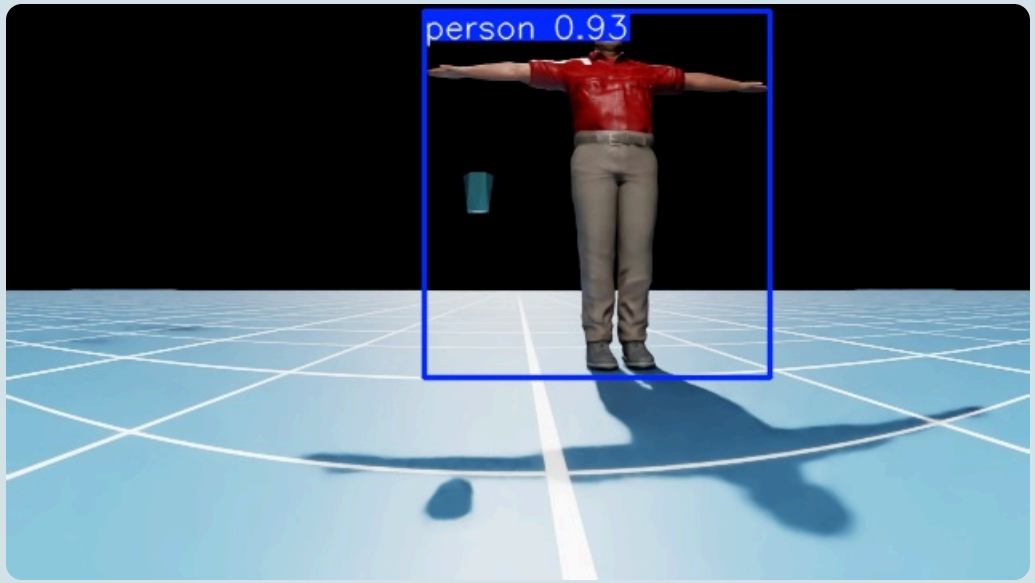


`/yolo/detections`
`/yolo/annotated_image`

No reliable 3D distance.

SCREENSHOT PLACEHOLDER

Isaac Sim / YOLO



Annotated image with live person detection

`Class · confidence · 2D bounding box`

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COMPLEMENTARY MEASUREMENTS

Why add LiDAR?



Camera

Classification and visual information

Pixel position and 2D boxes



LiDAR

3D XYZ points and accurate geometric range

Physical scene structure

Pixels (u,v) and LiDAR points (X,Y,Z) live in different coordinate frames. Establish their geometric relationship first.

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GEOMETRIC RELATIONSHIP

ROS TF connects the sensor frames

base_link

↙️ ↘️

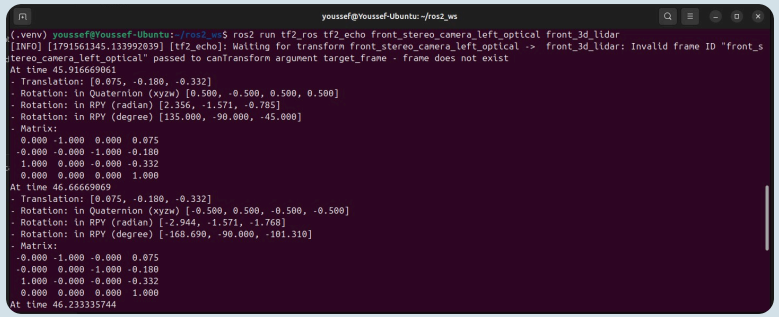
Camera
front_stereo_camera_left_optical

LiDAR
front_3d_lidar

Translation + rotation move LiDAR points into camera coordinates.

Static transforms: both sensors are rigidly mounted.

```
Validate: /tf_static
tf2_echo camera_frame lidar_frame
```



SCREENS

HOT PLACEHOLDER

RViz TF

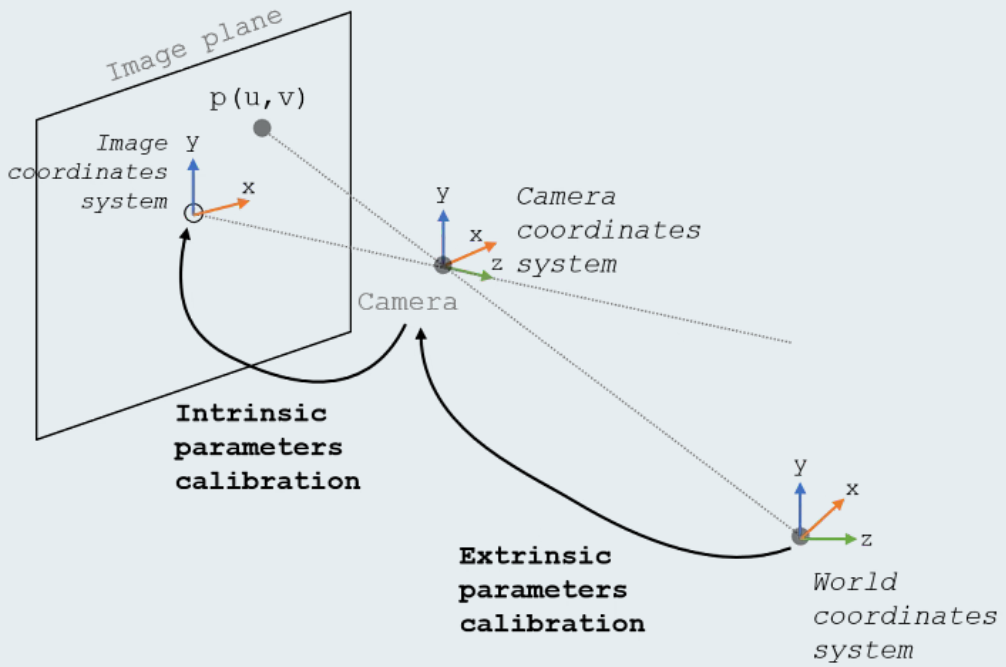
Camera and LiDAR frames connected through base_link

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PROJECTION

LiDAR-to-camera projection puts 3D points on the image

1. LiDAR XYZ
2. Transform into camera coordinates
3. Use CameraInfo calibration
4. Project onto RGB image
5. Obtain pixels (u,v)



Validate the RGB overlay: projected points should align with real surfaces.

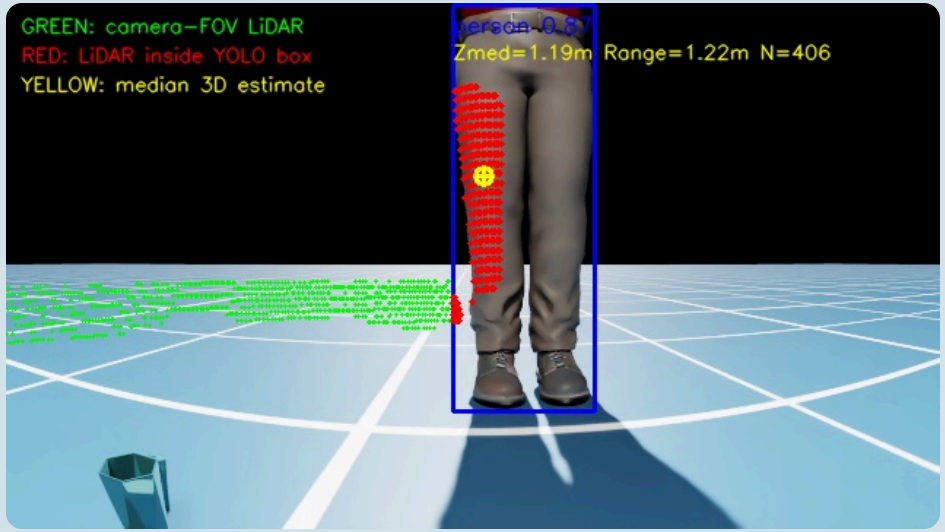
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ROI ASSOCIATION

YOLO boxes become ROIs for selecting LiDAR points

1. YOLO detection → bounding box / ROI
2. Projected LiDAR points
3. Select points inside ROI
4. Recover corresponding XYZ values
5. Estimate object position and distance

SCREENSHOT PLACEHOLDER



- Green dots: LiDAR points visible in camera
- Red dots: points inside YOLO ROI
- Highlighted point: estimated 3D position

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VALIDATION

Validating each stage independently

1 Camera

YOLO class, confidence, box

2 LiDAR

PointCloud2 in RViz

3 TF

Camera and LiDAR frames connected

4 Projection

Overlay on RGB image

5 ROI association

Selected points belong to the object

6 Distance

Compare with Isaac Sim reference

Fusion node returns LiDAR points and a distance estimate. Distance is still being validated: background and floor points may contaminate the box.

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DEPTH REFINEMENT

Improving distance accuracy by filtering the background

A YOLO rectangle also contains floor, wall behind and nearby objects.

01	02	03	04	05
ROI points	Depth distribution	Foreground filtering / clustering	Robust object cluster	Median / centroid XYZ

ROI shrinking · median filtering · depth histograms · nearest coherent depth cluster · DBSCAN

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SCREENSHOT PLACEHOLDER

Multiple people with overlapping detection boxes

Overlapping boxes and partial occlusion

DIFFICULT CASES

Multiple objects and difficult cases

- Separate LiDAR cluster per detection
- Sparse returns for distant or small targets
- Camera–LiDAR timestamp synchronization in moving scenes

Never assign the same LiDAR points to several objects.

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ROADMAP

From validated projection to 3D object localization

Completed

- Isaac Sim + ROS 2
- RGB acquisition + YOLO detection
- LiDAR PointCloud2
- Camera–LiDAR TF
- LiDAR → camera transformation
- Initial ROI association

Current

- Visual validation of projected points
- Range check against Isaac Sim reference
- Background removal

Next

- Robust depth estimation + accurate XYZ
- Multi-object association
- Occlusion handling + synchronization
- RViz 3D markers
- Complex warehouse scenes

Target: camera identifies → LiDAR locates in 3D → fused perception supports autonomous navigation.

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